Scalability & Future Plan

Date	2July,2026
Team ID	
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines how the **OptiCrop** application can be enhanced to support a larger number of users, improve prediction performance, and introduce additional smart farming features in future releases.

Current System Limitations

S.No	Limitation	Impact	Priority
1	Single-server deployment	Limited support for many concurrent users	Medium
2	No user login or prediction history	User data cannot be stored or tracked	Medium
3	Static crop information	Does not include live weather or soil updates	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports multiple users on a single Flask server	Deploy using cloud auto-scaling and load balancing
2	Data Storage	Uses local files and model storage	Integrate PostgreSQL or cloud database
3	Performance	Synchronous prediction requests	Add caching and optimize model inference
4	Data Integration	Uses static input values	Integrate real-time weather and soil APIs

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase-2	Weather API integration for real-time recommendations	Future Release	More accurate crop recommendations
Phase-3	Fertilizer recommendation and crop yield prediction	Future Release	Better farming decisions
Phase-4	User login, prediction history, and dashboard	Future Release	Personalized user experience
Phase-5	Mobile application and multilingual support	Future Release	Improved accessibility for farmers